

1	BASS/2025/I/06
	The curve is such that $\frac{dy}{dx} = 6e^{5x} - 3e^{-2x} + ax$, where a is a constant. The curve intersects the y-axis at $P(0,1)$ and $\frac{d^2y}{dx^2} = 40$ at point P. Find the equation of the curve. [7]
2	BASS/2025/II/06
	A curve has the equation $y = 2\sqrt{x} \ln(2x)$ where $x \geq 1$. (a) Show that $\frac{dy}{dx} = \frac{2}{\sqrt{x}} + \frac{\ln 2x}{\sqrt{x}}$. [3] (b) Find the rate of change of x at the instant when $x = 1$, given that y is changing at the rate of 5 units per second at this instant. [2] (c) Find $\int \frac{3 \ln 2x}{\sqrt{x}} dx$. [3]
3	ACSBR/2025/II/08
	(a) (i) Differentiate $2x \ln x - 6x$ with respect to x. [2] (ii) Hence find $\int_1^8 2 \ln x dx$, giving your answer in the form of $k + h \ln 2$, where h and k are constants. [4] (b) It is given that $\int_{-2}^0 f(x) dx = p$ and $\int_0^3 f(x) dx = q$. (i) Evaluate $\int_{-2}^3 f(x) dx$ in terms of p and q. [2] (ii) Find the value of m if $\int_0^{-2} [m - f(x)] dx = p - 10$. [3]

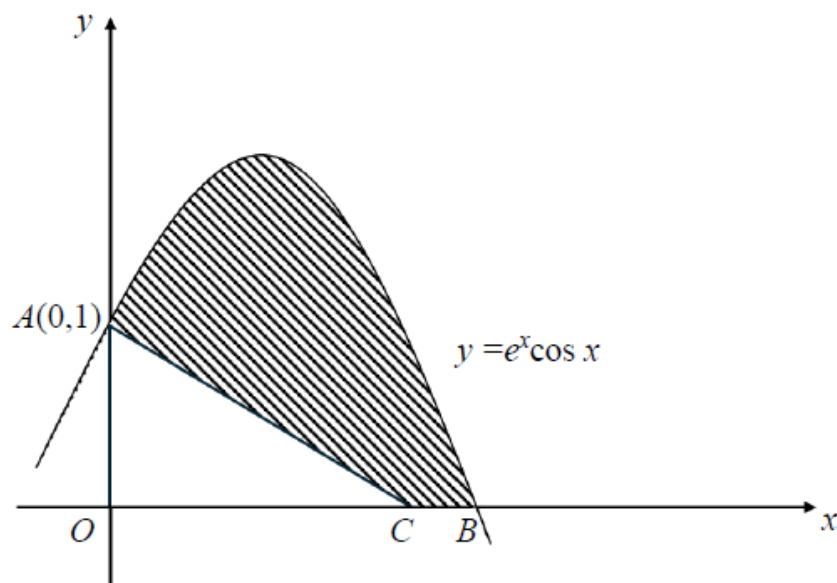
(a) Show that $\frac{d}{dx}[e^x(\cos x + \sin x)] = 2e^x \cos x$.

[2]

(b) Hence find $\int e^x \cos x \, dx$.

[2]

(c)



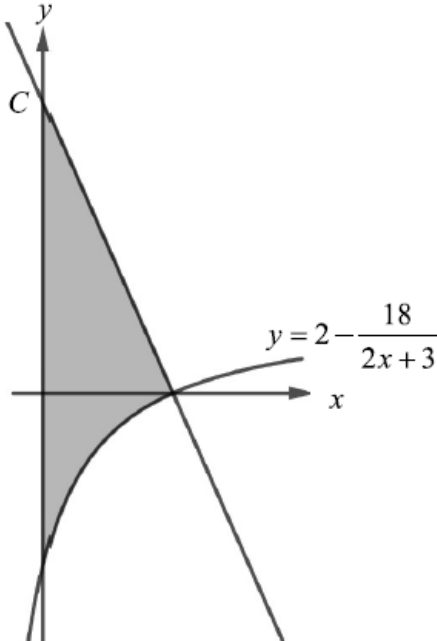
The diagram shows part of the curve $y = e^x \cos x$, cutting the x -axis at the point B . The normal to the curve at point $A(0, 1)$ cuts the x -axis at C .

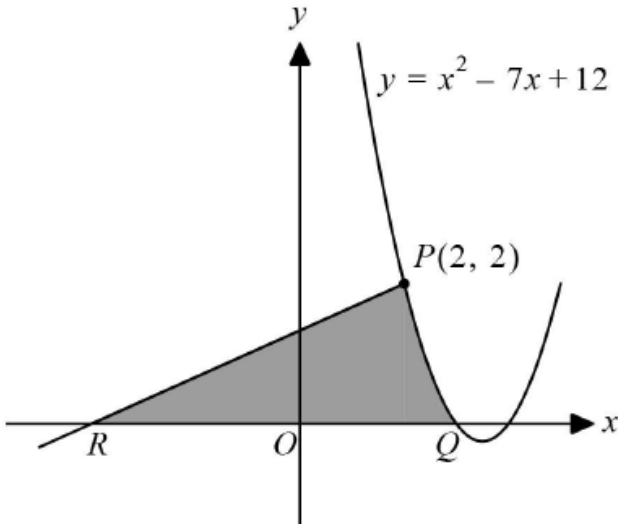
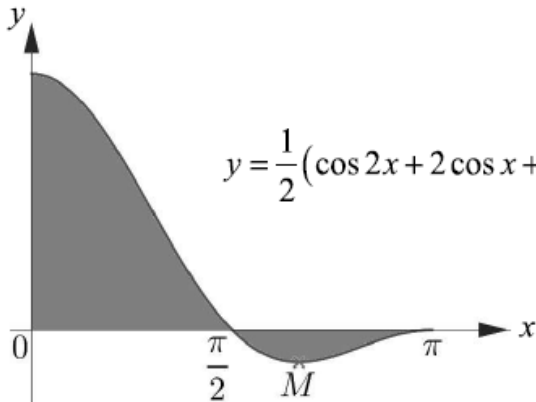
(i) Find the coordinates of B and of C .

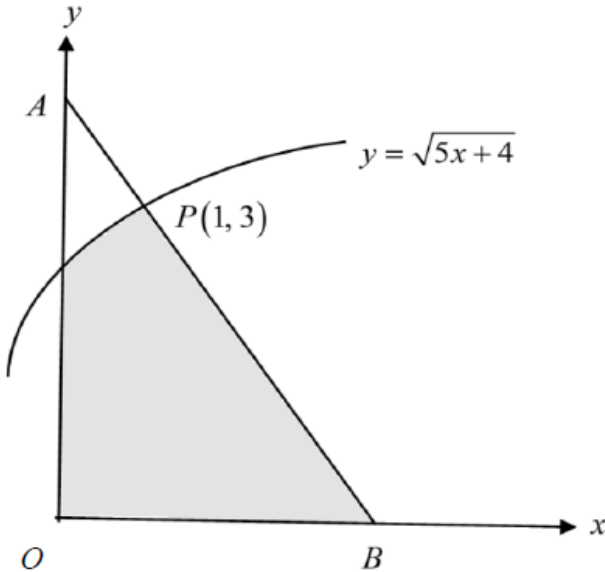
[5]

(ii) Find the area of the shaded region bounded by the curve, the line AC and the x -axis.

[3]

5	BASS/2025/II/10
	 The diagram shows part of the curve $y = 2 - \frac{18}{2x + 3}$, which crosses the x-axis at A and the y-axis at B. The normal to the curve at A crosses the y-axis at C. (a) Find the coordinates of C. [5] (b) Find the area of the shaded region $ABOC$ in terms of $a + b \ln 3$, where a and b are constants. [7]
6	CCHY/2025/I/2
	A container has a capacity of 840 cm^3 and is initially filled completely with water. The volume, $V \text{ cm}^3$, of water in the container is given by $V = h^2 + 2h$, where $h \text{ cm}$ is the height of the water level in the container. Due to leakage at the bottom of the container, the height of the water level in the container decreases at a rate of $\frac{5t}{3} \text{ cm/s}$, where t is the time in seconds. (a) Find the initial height of the water level in the container. [2] (b) Show that the height of the water level h can be expressed as $h = -\frac{5t^2}{6} + 28$. [2] (c) Find the rate of decrease of volume when $t = 3$. [5]
7	CCHY/2025/I/3
	(a) Find $\frac{d}{dx} \left(x e^{\frac{1}{2}x} \right)$. [2] (b) Hence evaluate $\int_0^4 x e^{\frac{1}{2}x} dx$, leaving your answer in the form $k(e^2 + 1)$, where k is a constant to be found. [5]

8	CCHY/2025/I/11
	(i) Express $\frac{11x+12}{x^2(x+4)}$ in partial fractions. [5] (ii) Hence find $\int \frac{11x+12}{x^2(x+4)} dx$. [2]
9	CCHY/2025/I/12
	The diagram shows part of the curve $y = x^2 - 7x + 12$, cutting the x-axis at Q. The normal to the curve at $P(2, 2)$ meets the x-axis at R.  Show that the area of the shaded region bounded by the x-axis, the line PR and the curve is [8]
10	CCHY/2025/II/7
	The diagram shows the curve $y = \frac{1}{2}(\cos 2x + 2 \cos x + 1)$ for $0 \leq x \leq \pi$ radians.  The point M is the minimum point of the curve, where the x-coordinate of M lies in the interval $\frac{\pi}{2} < x < \pi$. (a) Find the exact coordinates of M. [5] (b) Find the exact total area of the shaded regions bounded by the curve and the axes. [4]

11	GMS/2025/I/3
	(a) Differentiate $x\sqrt{2x-1}$ with respect to x. [2] (b) Hence find $\int \frac{x}{\sqrt{2x-1}} dx$. [3]
12	GMS/2025/I/12
	(a) Express $\frac{x^2-4x+4}{x^2(x+2)}$ as the sum of 3 partial fractions. [4] (b) Use your answer to part (a) to show that $\int_1^2 \frac{x^2-4x+4}{x^2(x+2)} dx = \ln\left(\frac{32}{81}\right) + 1$. [3]
13	GMS/2025/II/9
	 In the diagram above, triangle OAB represents a park. The development of a new road represented by the curve $y = \sqrt{5x+4}$, will result in the park being reduced to the shaded region shown. The side AB is a normal to the curve at the point $P(1, 3)$. (a) Find the equation of the line AB. [4] (b) Find the length of OA. [1] (c) Show that the development of the new road will reduce the park to approximately 85.5 % of its original size. [6]
14	ACSBR/2025/I/08
	A curve is such that $\frac{d^2y}{dx^2} = 2\sin x - 3\cos 2x$. The curve passes through the point $Q\left(\frac{\pi}{2}, 5\right)$ and the gradient of the curve at Q is 1. Find the equation of the curve. [6]

1

$$\frac{dy}{dx} = 6e^{5x} - 3e^{-2x} + ax$$

$$\frac{d^2y}{dx^2} = 30e^{5x} + 6e^{-2x} + a$$

Sub $x = 0$ at point P

$$\frac{d^2y}{dx^2} \bigg|_{x=0} = 30e^{5(0)} + 6e^{-2(0)} + a$$

$$30e^{5(0)} + 6e^{-2(0)} + a = 40$$

$$30 + 6 + a = 40$$

$$a = 4$$

$$\frac{dy}{dx} = 6e^{5x} - 3e^{-2x} + 4x$$

$$y = \int 6e^{5x} - 3e^{-2x} + 4x \, dx$$

$$y = \frac{6}{5}e^{5x} + \frac{3}{2}e^{-2x} + 2x^2 + C$$

At point P (0,1)

$$1 = \frac{6}{5}e^{5(0)} + \frac{3}{2}e^{-2(0)} + 2(0)^2 + C$$

$$C = \frac{-17}{10}$$

$$y = \frac{6}{5}e^{5x} + \frac{3}{2}e^{-2x} + 2x^2 - \frac{17}{10}$$

2	a	$\frac{dy}{dx} = (2\sqrt{x}) \frac{d(\ln 2x)}{dx} + (\ln 2x) \frac{d(2\sqrt{x})}{dx}$ $= (2\sqrt{x}) \left(\frac{1}{2x} \right) (2) + (\ln 2x) \left(\frac{1}{2} (2) x^{-\frac{1}{2}} \right)$ $= \frac{2}{\sqrt{x}} + \frac{\ln 2x}{\sqrt{x}} \text{ (shown)}$
	b	$\frac{dy}{dx} \Big _{x=1} = \frac{2}{\sqrt{1}} + \frac{\ln 2(1)}{\sqrt{1}}$ $= 2 + \ln 2$ $\frac{dx}{dt} = \frac{dy}{dt} \div \frac{dy}{dx}$ $\frac{dx}{dt} \Big _{x=1} = 5 \div (2 + \ln 2)$ $= \frac{5}{2 + \ln 2} \text{ or } 1.86 \text{ units/s}$
	c	$\therefore \int \frac{2}{\sqrt{x}} + \frac{\ln 2x}{\sqrt{x}} dx = (2\sqrt{x})(\ln 2x) + c_1$ $\therefore \int \frac{2}{\sqrt{x}} dx + \int \frac{\ln 2x}{\sqrt{x}} dx = (2\sqrt{x})(\ln 2x) + c_1$ $\therefore \int \frac{\ln 2x}{\sqrt{x}} dx = (2\sqrt{x})(\ln 2x) - \int \frac{2}{\sqrt{x}} dx + c_1$ $\therefore \int \frac{3 \ln 2x}{\sqrt{x}} dx = (6\sqrt{x})(\ln 2x) - \int \frac{6}{\sqrt{x}} dx + c_2$ $\therefore \int \frac{3 \ln 2x}{\sqrt{x}} dx = (6\sqrt{x})(\ln 2x) - \frac{6\sqrt{x}}{0.5} + c_3$ $\therefore \int \frac{3 \ln 2x}{\sqrt{x}} dx = (6\sqrt{x})(\ln 2x) - 12\sqrt{x} + c$

3	(a)(i)	$2x \frac{1}{x} + 2(\ln x - 3)$ $= 2 + 2 \ln x - 6$ $= 2 \ln x - 4$
	(a)(ii)	$\int_1^8 2 \ln x - 4 \, dx = [2x \ln x - 6x]_1^8$ $\int_1^8 2 \ln x \, dx - [4x]_1^8 = [16 \ln 8 - 48 - 2 \ln 1 + 6]$ $\int_1^8 2 \ln x \, dx - 28 = 16 \ln 8 - 42$ $\int_1^8 2 \ln x \, dx = 16 \ln 2^3 - 14$ $\int_1^8 2 \ln x \, dx = 48 \ln 2 - 14$ $= -14 + 48 \ln 2$
	(b)(i)	$\int_2^3 f(x) = \int_{-2}^0 f(x) \, dx + \int_0^3 f(x) \, dx$ $= p + q$
	(b)(ii)	$m[x]_0^{-2} + \int_{-2}^0 f(x) \, dx = p - 10$ $-2m + p = p - 10$ $2m = 10$ $m = 5$

$$\begin{aligned}
 \frac{dy}{dx} &= e^x(-\sin x + \cos x) + e^x(\cos x + \sin x) \\
 (a) \quad &= e^x(-\sin x + \cos x + \cos x + \sin x) \\
 &= e^x(2\cos x) \\
 &= 2e^x \cos x
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad &2 \int e^x \cos x \, dx = e^x(\cos x + \sin x) + c \\
 &\int e^x \cos x \, dx = \frac{1}{2} e^x(\cos x + \sin x) + c_1 \\
 &\text{at } B, y = 0 \\
 &e^x \cos x = 0 \\
 &\cos x = 0 \\
 &x = \frac{\pi}{2} \\
 &B\left(\frac{\pi}{2}, 0\right) \\
 &\frac{dy}{dx} = e^x(-\sin x) + e^x(\cos x) \\
 &\text{at } x = 0, \frac{dy}{dx} = \cos 0 - \sin 0 = 1 \\
 &\text{gradient of normal} = -1 \\
 &\text{equation of normal} \\
 &y - 1 = -(x - 0) \\
 &y = -x + 1 \\
 &\text{at } C, y = 0, x = 1 \\
 &C(1, 0)
 \end{aligned}$$

$$\begin{aligned}
 c \text{ (ii)} \quad &\int_0^{\frac{\pi}{2}} e^x \cos x \, dx - \frac{1}{2}(1)(1) \\
 &= \left[\frac{1}{2} e^x(\cos x + \sin x) \right]_0^{\frac{\pi}{2}} - \frac{1}{2} \\
 &= \frac{1}{2} \left[e^{\frac{\pi}{2}} - 1 \right] - \frac{1}{2} \\
 &= 1.41 \text{ sq units}
 \end{aligned}$$

5

a

$$y = 2 - \frac{18}{2x+3}$$

$$\frac{dy}{dx} = -18(-1)(2)(2x+3)^{-2}$$

$$= \frac{36}{(2x+3)^2}$$

x - axis, $y = 0$,

$$2 - \frac{18}{2x+3} = 0$$

$$2x+3 = 9$$

$$x = 3$$

$$\therefore A(3, 0)$$

$$\left. \frac{dy}{dx} \right|_{x=3} = \frac{36}{9^2}$$

$$= \frac{4}{9}$$

Eqn. of line AC :

$$y - 0 = -\frac{9}{4}(x - 3)$$

$$y = -\frac{9}{4}x + \frac{27}{4}$$

At y -axis,

$$y = 0 + \frac{27}{4}$$

$$\therefore C\left(0, 6\frac{3}{4}\right)$$

b

Area of shaded region $ABOC$

$$= \text{Area of } \triangle AOC - \int_0^3 \left(2 - \frac{18}{2x+3}\right) dx$$

$$= \frac{1}{2}(3)\left(\frac{27}{4}\right) - [2x - 9\ln(2x+3)]_0^3$$

$$= \frac{81}{8} - [6 - 9\ln 9 + 9\ln 3]$$

$$= 10\frac{1}{8} - 6 + 9\ln 9 - 9\ln 3$$

$$= \left(4\frac{1}{8} + 9\ln 3\right) \text{ units}^2 \text{ or } \left(\frac{33}{8} + 9\ln 3\right) \text{ units}^2 \text{ or } (4.125 + 9\ln 3) \text{ units}^2$$

6

(a)

$$\text{at } t = 0, V = 840$$

$$h^2 + 2h = 840$$

$$h^2 + 2h - 840 = 0$$

$$(h - 28)(h + 30) = 0$$

$$h = 28 \quad \text{or} \quad h = -30 \text{ (rejected)}$$

$$\therefore \text{Initial Height} = 28 \text{ cm}$$

(b)

$$h = \int -\frac{5t}{3} dt$$

$$= -\frac{5t^2}{6} + c$$

$$\text{when } t = 0, h = 28,$$

$$28 = 0 + c$$

$$c = 28$$

$$\therefore h = -\frac{5t^2}{6} + 28 \text{ (shown)}$$

(c)

$$\frac{dV}{dh} = 2h + 2$$

$$\begin{aligned} \frac{dh}{dt} &= \frac{-5(3)}{3} \\ &= -5 \end{aligned}$$

$$\text{when } t = 3,$$

$$\begin{aligned} h &= \frac{-5(3)^2}{6} + 28 \\ &= 20.5 \end{aligned}$$

$$\begin{aligned} \frac{dV}{dh} &= 2(20.5) + 2 \\ &= 43 \end{aligned}$$

$$\begin{aligned} \frac{dV}{dt} &= \frac{dV}{dh} \times \frac{dh}{dt} \\ &= 43 \times (-5) \end{aligned}$$

$$= -215 \text{ cm}^3 / \text{s}$$

$$\text{Rate of decrease} = 215 \text{ cm}^3 / \text{s}$$

(a)

$$\frac{d}{dx} \left(x e^{\frac{1}{2}x} \right) = x e^{\frac{1}{2}x} \left(\frac{1}{2} \right) + e^{\frac{1}{2}x}$$

$$= \frac{1}{2} x e^{\frac{1}{2}x} + e^{\frac{1}{2}x}$$

$$\text{Accept } e^{\frac{1}{2}x} \left(\frac{1}{2}x + 1 \right)$$

(b)

$$\int_0^4 \frac{1}{2} x e^{\frac{1}{2}x} + e^{\frac{1}{2}x} dx = \left[x e^{\frac{1}{2}x} \right]_0^4$$

$$\int_0^4 x e^{\frac{1}{2}x} + 2e^{\frac{1}{2}x} dx = 2 \left[x e^{\frac{1}{2}x} \right]_0^4$$

$$\int_0^4 x e^{\frac{1}{2}x} dx = 2 \left[x e^{\frac{1}{2}x} \right]_0^4 - \int_0^4 2e^{\frac{1}{2}x} dx$$

$$= 2 \left[x e^{\frac{1}{2}x} \right]_0^4 - \left[4e^{\frac{1}{2}x} \right]_0^4$$

$$= 2[4e^2 - 0] - [4e^2 - 4]$$

$$= 8e^2 - 4e^2 + 4$$

$$= 4e^2 + 4$$

$$= 4(e^2 + 1)$$

8**(i)**

$$\frac{11x+12}{x^2(x+4)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+4}$$

$$11x+12 = Ax(x+4) + B(x+4) + Cx^2$$

when $x = 0$,

$$12 = 0 + 4B + 0$$

$$B = 3$$

when $x = -4$,

$$-32 = 0 + 0 + 16C$$

$$C = -2$$

when $x = 1$,

$$23 = 5A + 15 - 2$$

$$A = 2$$

$$\therefore \frac{11x+12}{x^2(x+4)} = \frac{2}{x} + \frac{3}{x^2} - \frac{2}{x+4}$$

(ii)

$$\int \frac{11x+12}{x^2(x+4)} dx = \int \frac{2}{x} + \frac{3}{x^2} - \frac{2}{x+4} dx$$

$$= 2 \ln x - \frac{3}{x} - 2 \ln(x+4) + c$$

$$\frac{dy}{dx} = 2x - 7$$

when $x = 2$,

$$\begin{aligned}\frac{dy}{dx} &= 2(2) - 7 \\ &= -3\end{aligned}$$

$$\text{Gradient of normal} = \frac{1}{3}$$

$$y = \frac{1}{3}x + c$$

at $(2, 2)$

$$2 = \frac{2}{3} + c$$

$$c = \frac{4}{3}$$

$$\text{Equation of normal: } y = \frac{1}{3}x + \frac{4}{3}$$

when $y = 0$,

$$-\frac{1}{3}x = \frac{4}{3}$$

$$x = -4$$

$$R = (-4, 0)$$

when $y = 0$,

$$x^2 - 7x + 12 = 0$$

$$(x-3)(x-4) = 0$$

$$x = 3 \text{ or } x = 4$$

$$Q = (3, 0)$$

$$\text{Area of shaded region} = \frac{1}{2}(6)(2) + \int_2^3 x^2 - 7x + 12 \, dx$$

$$= 6 + \left[\frac{x^3}{3} - \frac{7}{2}x^2 + 12x \right]_2^3$$

$$= 6 + \left[\left(9 - \frac{63}{2} + 36 \right) - \left(\frac{8}{3} - 14 + 24 \right) \right]$$

$$= 6\frac{5}{6} \text{ units}^2 \text{ (shown)}$$

(a)

$$y = \frac{1}{2}(\cos 2x + 2\cos x + 1)$$

$$\frac{dy}{dx} = \frac{1}{2}(-2\sin 2x - 2\sin x)$$

$$= -\sin 2x - \sin x$$

$$\text{At min pt, } \frac{dy}{dx} = 0.$$

$$-\sin 2x - \sin x = 0$$

$$-2\sin x \cos x - \sin x = 0$$

$$-\sin x(2\cos x + 1) = 0$$

$$-\sin x = 0 \quad \text{or} \quad \cos x = -\frac{1}{2}$$

$$x = 0 \text{ or } \frac{\pi}{2} \text{ (rej)}$$

$$x = \pi - \frac{\pi}{3} \quad \text{or} \quad \pi + \frac{\pi}{3} \text{ (rej)}$$

$$y = -\frac{1}{4}$$

$$\therefore M\left(\frac{2\pi}{3}, -\frac{1}{4}\right)$$

(b)

$$\text{area} = \int_0^{\frac{\pi}{2}} \frac{1}{2}(\cos 2x + 2\cos x + 1) \, dx - \int_{\frac{\pi}{2}}^{\pi} \frac{1}{2}(\cos 2x + 2\cos x + 1) \, dx$$

$$= \frac{1}{2} \left[\frac{\sin 2x}{2} + 2\sin x + x \right]_0^{\frac{\pi}{2}} - \frac{1}{2} \left[\frac{\sin 2x}{2} + 2\sin x + x \right]_{\frac{\pi}{2}}^{\pi}$$

$$= \left[\left(0 + 1 + \frac{\pi}{4} \right) - (0 + 0 + 0) \right] - \left[\left(0 + 0 + \frac{\pi}{2} \right) - \left(0 + 1 + \frac{\pi}{4} \right) \right]$$

$$= 2 \text{ units}^2$$

11**(a)**

$$\begin{aligned}\frac{d}{dx}(x\sqrt{2x-1}) &= x\left(\frac{1}{2}\right)(2x-1)^{-\frac{1}{2}}(2) + (2x-1)^{\frac{1}{2}} \\ &= (2x-1)^{-\frac{1}{2}}[x + 2x-1] \\ &= \frac{3x-1}{\sqrt{2x-1}}\end{aligned}$$

(b)

$$\begin{aligned}\int \frac{3x-1}{\sqrt{2x-1}} dx &= x\sqrt{2x-1} \\ 3\int \frac{x}{\sqrt{2x-1}} dx - \int \frac{1}{\sqrt{2x-1}} dx &= x\sqrt{2x-1} \\ 3\int \frac{x}{\sqrt{2x-1}} dx &= x\sqrt{2x-1} + \frac{(2x-1)^{\frac{1}{2}}}{\left(\frac{1}{2}\right)^2} + c \\ \int \frac{x}{\sqrt{2x-1}} dx &= \frac{1}{3}\left[x\sqrt{2x-1} + \sqrt{2x-1}\right] + c \\ &= \frac{1}{3}(\sqrt{2x-1})(x+1) + c\end{aligned}$$

(a)

$$\text{Let } \frac{x^2 - 4x + 4}{x^2(x+2)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+2}$$

$$x^2 - 4x + 4 = Ax(x+2) + B(x+2) + Cx^2$$

$$\text{Let } x = 0,$$

$$4 = 2B \Rightarrow B = 2$$

$$\text{Let } x = -2,$$

$$16 = 4C \Rightarrow C = 4$$

Comparing x^2 term,

$$A + C = 1 \Rightarrow A = -3$$

$$\frac{x^2 - 4x + 4}{x^2(x+2)} = -\frac{3}{x} + \frac{2}{x^2} + \frac{4}{x+2}$$

(b)

$$\int_1^2 \frac{x^2 - 4x + 4}{x^2(x+2)} dx = \int_1^2 \left(-\frac{3}{x} + \frac{2}{x^2} + \frac{4}{x+2} \right) dx$$

$$= \left[-3 \ln x - \frac{2}{x} + 4 \ln(x+2) \right]_1^2$$

$$= \left[-3 \ln 2 - \frac{2}{2} + 4 \ln 4 \right] - \left[-3 \ln 1 - \frac{2}{1} + 4 \ln 3 \right]$$

$$= -3 \ln 2 + 4 \ln 4 - 4 \ln 3 - 1 + 2$$

$$= -\ln 2^3 + \ln 4^4 - \ln 3^4 - 1 + 2$$

$$= -\ln 8 + \ln 256 - \ln 81 + 1$$

$$= \ln \frac{256}{8(81)} + 1$$

$$= \ln \frac{32}{81} + 1$$

(a)

$$y = (5x + 4)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{1}{2}(5x + 4)^{-\frac{1}{2}}(5)$$

$$= \frac{5}{2\sqrt{5x + 4}}$$

At point P , $x = 1$, $\frac{dy}{dx} = \frac{5}{6}$

Gradient of normal $= -\frac{6}{5}$

Equation of AB : $y - 3 = -\frac{6}{5}(x - 1)$

$$y = -\frac{6}{5}x + \frac{21}{5}$$

(b)

At point B , $y = 0$

$$y = -\frac{6}{5}x + \frac{21}{5}$$

$$-\frac{6}{5}x + \frac{21}{5} = 0$$

$$x = \frac{21}{6}$$

$$= 3.5 \quad (\text{Shown})$$

At point A , $x = 0$,

$$y = \frac{21}{5}$$

$$= 4.2$$

(c)

Area of $\triangle OAB = \frac{1}{2}(3.5)(4.2)$

$$= 7.35$$

Shaded area $= \int_0^1 (\sqrt{5x + 4}) dx + \frac{1}{2}(2.5)(3)$

$$= \left[\frac{(5x + 4)^{\frac{3}{2}}}{\frac{3}{2}(5)} \right]_0^1 + 3.75$$

$$= \frac{2}{15}(3)^3 - \frac{2}{15}(2)^3 + 3.75$$

$$= \frac{377}{60}$$

Percentage $= \left(\frac{377}{60} \div 7.35 \right) \times 100$

$$= 85.5 \%$$

14

$$\frac{dy}{dx} = \int 2 \sin x - 3 \cos 2x \, dx$$

$$= -2 \cos x - \frac{3 \sin 2x}{2} + c_1$$

$$x = \frac{\pi}{2}, \frac{dy}{dx} = 1$$

$$c_1 = 1$$

$$y = \int -2 \cos x - \frac{3 \sin 2x}{2} + 1 \, dx$$

$$= -2 \sin x + \frac{3 \cos 2x}{2} + x + c_2$$

$$x = \frac{\pi}{2}, y = 5$$

$$-2 - \frac{3}{4} + \frac{\pi}{2} + c_2 = 5$$

$$c_2 = \frac{31 - 2\pi}{4}$$

$$y = -2 \sin x + \frac{3}{2} \cos 2x + x + \frac{31 - 2\pi}{4}$$